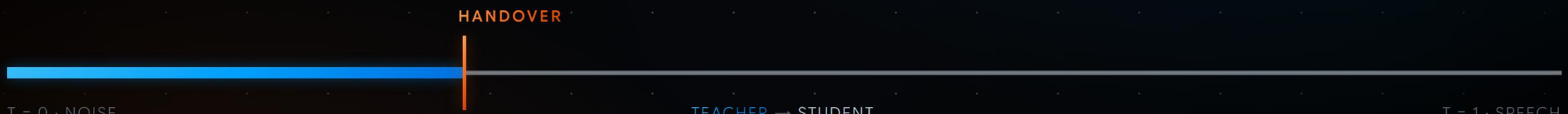
A RISK-CALIBRATED ADAPTIVE HANDOVER METHOD FOR TEACHER-GUIDED FLOW-MATCHING TEXT-TO-SPEECH INFERENCE



(Limpahan, M.V., Locsin, R., Sajol R. J. 2026. A Risk-Calibrated Adaptive Handover Method for Teacher-Guided Flow-Matching Text-to-Speech Inference)

THE WALKTHROUGH

THREE CHAPTERS, ONE DECISION

I INTRODUCTION

1.1 Background of the study
1.2 Problem statement
1.3 Objectives
1.4 Significance
1.5 Scope and limitations

II REVIEW OF LITERATURE

2.1 Flow matching and its cost
2.3 Teacher-guided sampling
2.4 Input-adaptive computation
2.5 Statistical risk control
2.8 Research gap

III METHODOLOGY

3.1 System architecture
3.5 Signals, policy and properties
3.6 Calibration with a guarantee
3.7 Benchmarking and robustness
App. Expected outputs

BACKGROUND OF THE STUDY

SPEECH, GENERATED FROM NOISE

A **flow-matching** model learns a velocity field that moves random noise toward a mel-spectrogram along a nearly straight path.

$$dx/dt = v_\vartheta(x,\ t), \quad t \in [0,\ 1]$$

generation = integrate the field step by step

Every step is one full network evaluation. F5-TTS, our teacher, spaces its steps with sway sampling.



SOURCES Lipman et al. (2023), *Flow matching for generative modeling*, ICLR · Liu et al. (2023), ICLR · Y. Chen et al. (2025), *F5-TTS*, ACL

BACKGROUND OF THE STUDY

EVERY STEP IS A NETWORK CALL

GUIDANCE DOUBLES IT

$$v = v_u + w(v_c - v_u)$$

Conditional + unconditional prediction: **2 forward passes** per guided teacher step.

DISTILLATION CUTS STEPS

4 steps

DMOSpeech distilled the teacher into a **few-step student**, 128 → 4.

…BUT LOSES DIVERSITY

The student's prosody is less varied (**mode shrinkage**). DMOSpeech 2 lets the teacher run the first steps and the student finish.

Released sampler, switching time 0.25 ■ teacher 18 ■ student 3



21 forward passes for **every** utterance

SOURCES Ho & Salimans (2022), *Classifier-free diffusion guidance* · Li et al. (2025), *DMOSpeech*, ICML · Li et al. (2026), *DMOSpeech 2*, AAAI

PROBLEM STATEMENT

ONE HANDOVER FOR EVERY INPUT



Too late wastes guided passes. Too early may degrade speech.
Does every utterance need 18 teacher passes?

1.2.1 **Uncharacterized effects.** How the switching time changes WER, SIM, UTMOS and cost per utterance is unmeasured.

1.2.2 **The handover boundary.** Can inference-time signals pick an earlier switch without losing quality?

1.2.3 **No reliability guarantee.** Adaptive selection alone says nothing about the rate of added failures.

1.2.4 **Application constraints.** Faster synthesis is not evidence of benefit to autistic users.

SOURCES Li et al. (2026), *DMOSpeech 2*, AAAI · Ho & Salimans (2022) · Angelopoulos et al. (2025), *Learn then test*, Ann. Appl. Stat.

OBJECTIVES OF THE STUDY

CHOOSE THE SWITCH PER UTTERANCE, WITH A BOUND ON FAILURES

1
CHARACTERIZE

Measure how WER, SIM, UTMOS, forward passes and runtime vary across candidate switching times and inputs.

→ QUALITY–COST MAP

2
FORMULATE

Pose selection as constrained cost minimization; design a training-free policy with no extra network evaluations; prove its properties.

→ POLICY + 4 PROPOSITIONS

3
CALIBRATE

Use Learn-Then-Test to keep added-failure risk within a tolerance α at confidence $1 - \delta$; check it on held-out speakers.

→ CERTIFIED THRESHOLD $\hat{\lambda}$

4
COMPARE

Benchmark SNR-only, stabilization-only and combined variants against fixed switches and an oracle, clean and noisy.

→ DOES EACH SIGNAL ADD VALUE?

FROM Proposal, Section 1.3 · all pretrained weights frozen; training-free

SIGNIFICANCE · SCOPE AND LIMITATIONS

ONE VARIABLE. EVERYTHING ELSE FROZEN.

- IN **Only the switching time varies.** Student restart level stays at the released default.
- IN Teacher, student, duration predictor and vocoder **stay frozen**.
- IN English zero-shot prompts from **Seed-TTS test-en**, clean and 15 dB noisy.
- IN All runs on an **NVIDIA Tesla T4** through Google Colab.
- OUT A per-utterance guarantee; the bound holds **on average** over inputs.
- OUT Building or testing an assistive app; clinical or learner outcomes.

WHO BENEFITS

SPEECH RESEARCHERS
A formulation, proofs and calibration for teacher-guided samplers in general.

TTS DEVELOPERS
Lower cost with a stated bound, no retraining.

ASSISTIVE-TECH DEVELOPERS
Cheaper, predictable synthesis for personalized voices.

FUTURE RESEARCHERS
A reusable protocol: disjoint calibration, fixed failure rule, exact cost.

SOURCES Anastassiou et al. (2024), *Seed-TTS* · Ardila et al. (2020), *Common Voice*, LREC · Muharib et al. (2024), JADD · Therrien et al. (2025), AAC

REVIEW OF RELATED LITERATURE

THE CHAIN OF IDEAS



FLOW MATCHING

GIVES

A trajectory, and a free estimate of the final output at every step

LEAVES OPEN

How a handover behaves

DISTILLATION

GIVES

A cheap few-step student

LEAVES OPEN

Lost diversity; one config for all inputs

TEACHER-GUIDED SAMPLING

GIVES

Diversity back; the handover as a design choice

LEAVES OPEN

Switch fixed for every utterance

ADAPTIVE COMPUTATION

GIVES

Spend effort only where it is needed

LEAVES OPEN

Mostly no bound on quality loss

STATISTICAL RISK CONTROL

GIVES

A threshold with a guarantee on new inputs

LEAVES OPEN

Not yet applied to TTS handover

THIS STUDY

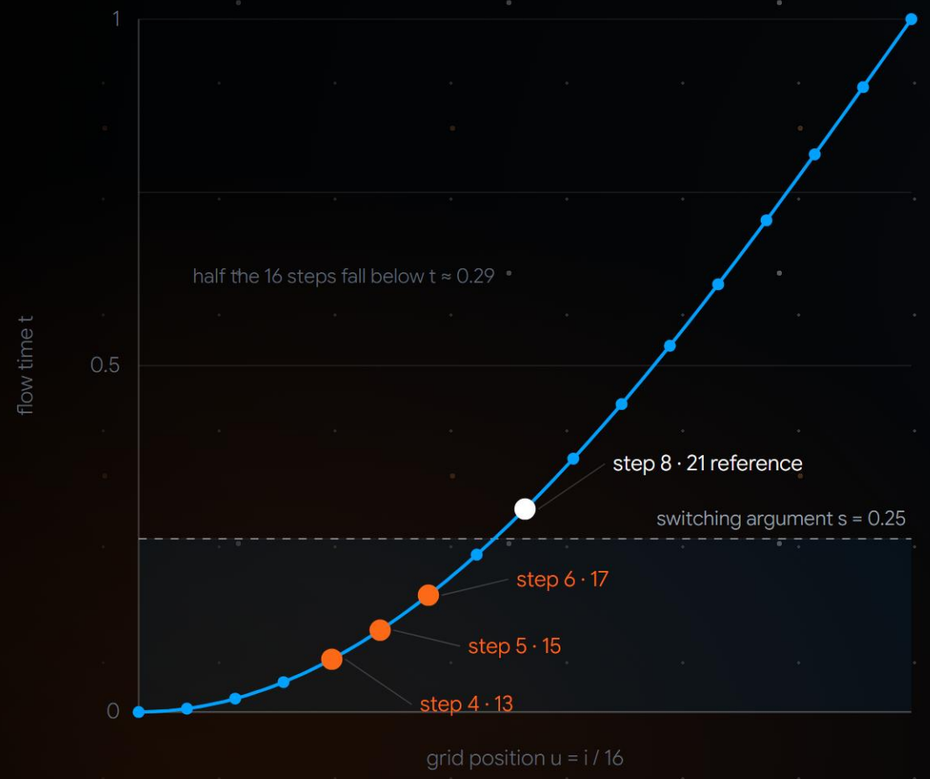
COMBINES

Per-utterance handover + calibrated bound, at no extra pass

SOURCES Lipman et al. (2023) · Yin et al. (2024) · Li et al. (2025, 2026) · Graves (2016) · Schuster et al. (2022), *CALM*, NeurIPS · Angelopoulos et al. (2025)

TEACHER-GUIDED SAMPLING AND RE-NOISING

WHERE THE SWITCH LIVES



Sway sampling packs teacher steps near $t = 0$. The released code stops at the **first grid point above** s. Cost: $C(s) = 2(k + 1) + 3$.



s = 0.07
step 4 · t = 0.076

13
−8 · 38%



s = 0.10
step 5 · t = 0.118

15
−6 · 29%



s = 0.15
step 6 · t = 0.169

17
−4 · 19%



s = 0.25
step 8 · reference

21
baseline

$$x_\tau = (1 - \tau)\,\varepsilon + \tau\,\tilde{x}, \quad \tau = 0.0761$$

student always re-noises the teacher's estimate to the same level

SOURCES Y. Chen et al. (2025), *F5-TTS*, ACL · Li et al. (2026), *DMOSpeech 2*, AAAI · Meng et al. (2022), *SDEdit*, ICLR · Xu et al. (2023), NeurIPS

INPUT-ADAPTIVE COMPUTATION

SPEND EFFORT WHERE IT IS NEEDED

METHOD	WHAT IS ADAPTED	DECISION SIGNAL	RULE	GUARANTEE
ACT Graves, 2016	Internal steps per input	Learned halting unit	Learned	NONE
BranchyNet Teerapittayanon et al., 2016	Exit layer	Classifier confidence	Tuned threshold	NONE
CALM Schuster et al., 2022	Exit layer per token	Confidence or saturation	Calibrated threshold	RISK CONTROL
Adaptive collaboration Starodubcev et al., 2024	Whether the teacher refines	Quality estimator on a full student sample	Tuned threshold	NONE
This study	Teacher-to-student switching time	Prompt SNR + endpoint stabilization	Calibrated threshold	RISK CONTROL (LTT)

SOURCES Graves (2016) · Teerapittayanon et al. (2016), ICPR · Schuster et al. (2022), NeurIPS · Starodubcev et al. (2024), CVPR · Mualem et al. (2026), *ARIA*

STATISTICAL RISK CONTROL AND CALIBRATION

A PROMISE YOU CAN CHECK

$$P(\ R(\hat{\lambda}) \leq \alpha\ ) \geq 1 - \delta$$

"At most 5% added failures, and this promise is wrong with probability at most 10%."

Learn-Then-Test turns each candidate threshold into a hypothesis test and needs no monotonicity. Testing in a **fixed order** keeps the overall error at δ.

INPUTS NEEDED · α = 0.05, δ = 0.10, ZERO FAILURES

$$(1-\alpha)^n \leq \delta \ \Rightarrow\ n \geq \ln \delta \,/\, \ln(1-\alpha)$$



The guarantee is **marginal** (on average over inputs) and needs calibration and test inputs from the same distribution.

SOURCES Angelopoulos & Bates (2023) · Angelopoulos et al. (2025), Ann. Appl. Stat. · Clopper & Pearson (1934) · Hoeffding (1963) · Bauer (1991)

SYNTHESIS AND RESEARCH GAP

NO METHOD COMBINES ALL THREE

REQUIREMENT 01

A per-utterance handover decision

made from signals the sampler already computes

REQUIREMENT 02

All networks stay as released

no training, no fine-tuning, no extra forward pass

REQUIREMENT 03

Added failures bounded with stated confidence

on new, unseen speakers

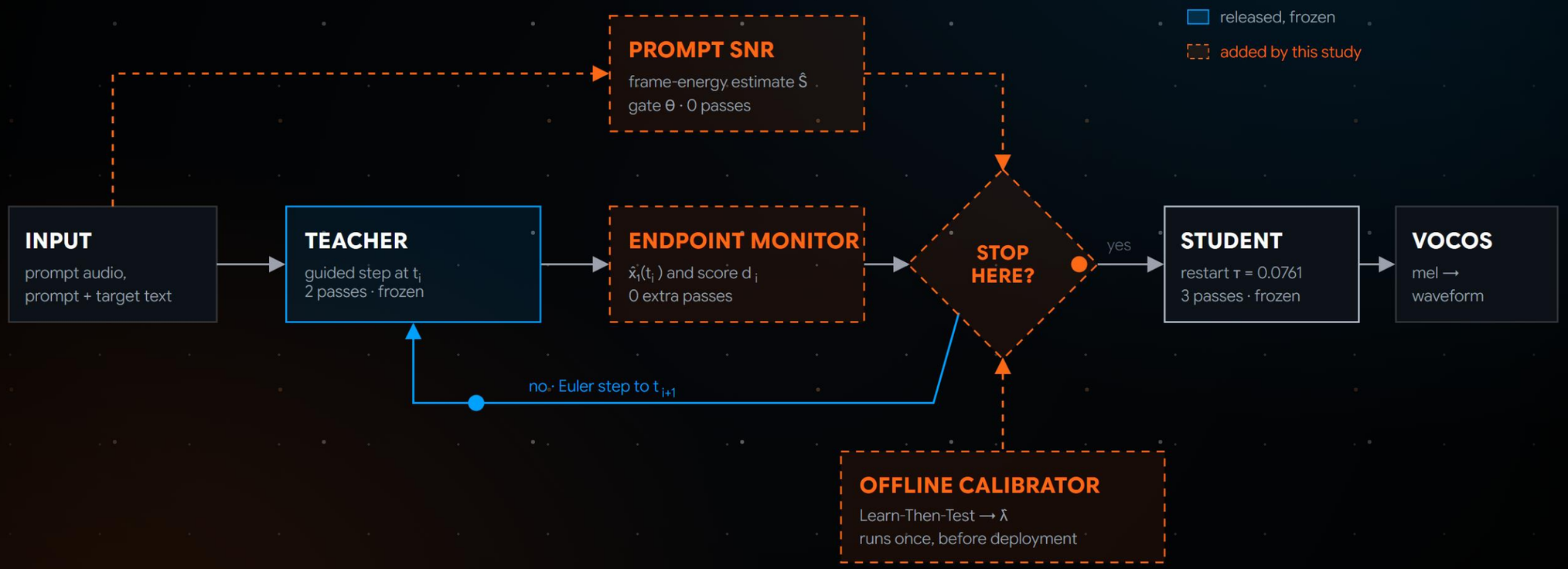
THIS STUDY

The gap concerns the combination of these elements within the sources examined; it does not imply that no other adaptive inference methods exist.

FROM Proposal, Section 2.8 and Table 2.2

SYSTEM ARCHITECTURE

A DECISION LAYER, NOT A NEW MODEL



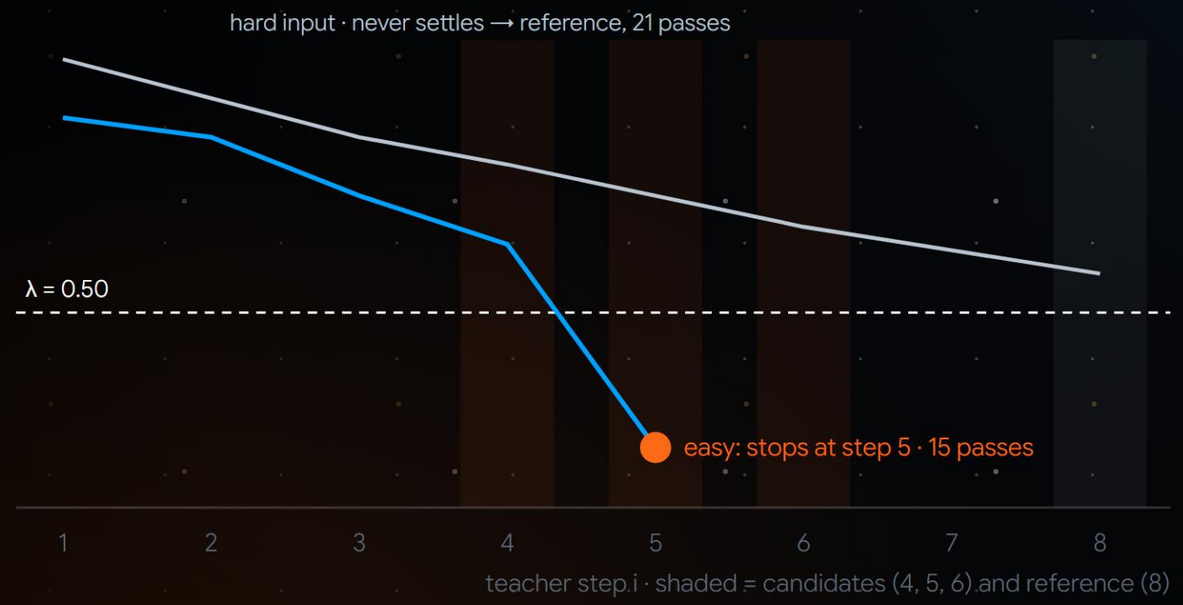
SOURCES Li et al. (2026), *DMOSpeech 2*, AAAI · Y. Chen et al. (2025), *F5-TTS* · Siuzdak (2024), *Vocos*, ICLR · Proposal Figure 3.2

DECISION SIGNALS

TWO SIGNALS, ZERO EXTRA PASSES

$$d_i = \| \hat{x}_1(t_i) - \hat{x}_1(t_{i-1}) \|_M \,/\, (\| \hat{x}_1(t_i) \|_M + \epsilon), \quad \hat{x}_1 = x_t + (1 - t)\, v$$

endpoint-stabilization score · small means the teacher's guess has settled



$$\hat{S} = 10 \log_{10}(\bar{E}_{\text{top}} \,/\, \bar{E}_{\text{bottom}})$$

prompt SNR, from frame energies · gate θ = 20 dB



Development diagnostics suggest the **SNR may carry most of the benefit**. The stabilization score is kept as a hypothesis to test, not an assumption.

SOURCES Lipman et al. (2024), *Flow matching guide and code* · Li et al. (2026) · Proposal Def. 3.2–3.3, Appendix A.4 and D.5 (illustrative values)

FORMULATION AND PROPERTIES

THE STOPPING RULE

$$\min_{\pi} \mathrm{E}[\ C(\pi(x))\ ] \quad \text{s.t.} \quad R(\pi) = \mathrm{E}[\ f(x, \pi(x))\ ] \le \alpha$$

cheapest policy whose added-failure risk stays under α

$$\pi_{\lambda}(x) = \text{first candidate } i_r \text{ with } d_{i_r} \le \lambda \text{ and } \hat{S} \ge \vartheta, \text{ else reference}$$

walks the teacher steps in order · never later than step 8

ADDED FAILURE · FIXED BEFORE CALIBRATION

$f = 1$ if WER > max(**12.5%**, $\mathrm{WER}_{\mathrm{ref}}$ + **1 pt**) or SIM < $\mathrm{SIM}_{\mathrm{ref}}$ − **0.02**, on the same seeds.

Setting λ = ∞ gives the SNR-only rule; θ = −∞ gives stabilization only. **One family, one calibration.**

PROP. 3.2

ZERO OVERHEAD

Signals reuse the velocity already computed. No extra forward pass.

PROP. 3.3

MONOTONE COST

A smaller λ never stops earlier, so cost rises as λ falls.

PROP. 3.4

SAFE FALLBACK

Never more than 21 passes. No certified λ → exact reference output.

PROP. 3.5

ORACLE BOUND

No zero-risk policy beats the cheapest non-failing candidate per input.

FROM Proposal Eq. 3-11, 3-12, 3-15 and Propositions 3.2–3.5 · WER: Radford et al. (2023) · SIM: S. Chen et al. (2022)

CALIBRATION WITH A RELIABILITY GUARANTEE

TEST THRESHOLDS IN A FIXED ORDER

SPEAKER-DISJOINT SPLITS · SEED-TTS TEST-EN · ONE UTTERANCE PER SPEAKER

Development	Calibration	Test
100	200	200

ILLUSTRATIVE · n = 100, α = 0.05, δ = 0.10 · p = P(Bin(n, α) ≤ failures)

g = 1 · conservative	g = 2	g = 3	g = 4 · aggressive
λ 0.40	**λ 0.50**	**λ 0.60**	**λ 0.70**
failures 0	failures 1	failures 3	failures 6
p 0.0059	p 0.0371	p 0.2578	p —
≤ Δ · CERTIFY	≤ Δ · CERTIFY	> Δ · STOP	NOT TESTED

λ̂ = 0.50

Study protocol: α = 0.05, δ = 0.10, grid 0.20 to 1.20 in steps of 0.05, fixed in advance.
Verify on the **held-out test split** and over **100 random re-splits**.

SOURCES Angelopoulos et al. (2025), *Learn then test* · Bauer (1991), Stat. Med. · Clopper & Pearson (1934) · Proposal Alg. 3.4, Appendix A.6

RESEARCH WORKFLOW · BENCHMARKING AND ROBUSTNESS

FIFTEEN STEPS, EACH WITH ITS OWN DATA

SETUP — ALL SPLITS
1 Speaker splits · 2 Clean + 15 dB noisy · 3 Check seed, fallback, pass count

OBJECTIVE 1 — DEVELOPMENT
4 One teacher run, all candidates · 5 Score + failures · 6 Quality–cost map, fix θ

OBJECTIVE 2 — NO DATA
7 Problem, policy, proofs

OBJECTIVE 3 — CALIBRATION, TEST
8 Fix α, δ, grid · 9 Binomial tests in order · 10 Lock $\hat{\lambda}$ or fall back · 11 Held-out + 100 re-splits

OBJECTIVE 4 — TEST
12 Run locked variants · 13 Paired comparisons · 14 Noise shift, n = 50/100/200 · 15 Report all, incl. negative

COMPARED METHODS

Fixed s ∈ {0.07, 0.10, 0.15, **0.25**} · SNR-only · SNR + fixed later step · stabilization only · **full policy** · cost-matched random · oracle

MEASURES

WER Whisper-large-v3 · **SIM** WavLM-Base+ SV · **UTMOS** · exact **forward passes** · runtime and RTF on the T4

STATISTICS

Wilcoxon signed-rank + Holm · exact McNemar · TOST for equivalence · Clopper–Pearson intervals · bootstrap CIs

SOURCES Radford et al. (2023) · S. Chen et al. (2022) · Saeki et al. (2022), *UTMOS* · Wilcoxon (1945) · Schuirmann (1987)

PROTOTYPE AND SAMPLE SIMULATION

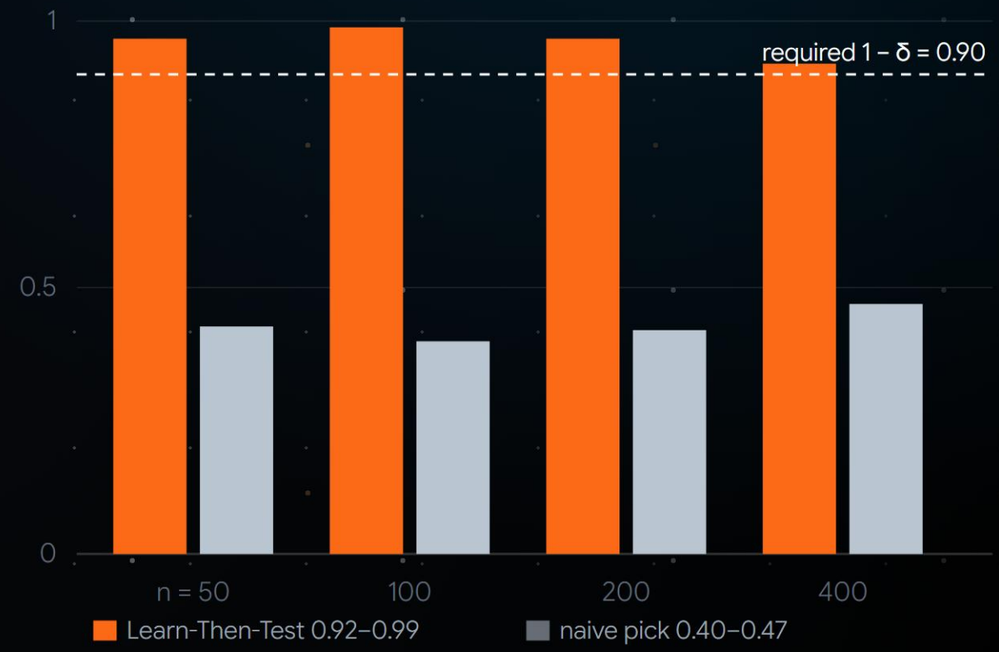
WHAT WE EXPECT TO SEE

PROTOTYPE · HANDOVER PER INPUT



Easy inputs save up to **38%**. Hard inputs cost **exactly the baseline**, never more.

SIMULATION · P(R ≤ α) OVER 1,000 RUNS



Simulated data, not experimental results. LTT used 15.9–18.2 passes on average vs 21 for the reference.

FROM Proposal Appendix B (Table B.1) and Appendix C (Figure C.1) · costs exact by Proposition 3.1

THE CONTRIBUTION

PER-UTTERANCE HANDOVER, WITH A CERTIFIED BOUND ON ADDED FAILURES

DECIDE

Stop the teacher once its guess settles and the prompt is clean. Zero extra passes.

GUARANTEE

$\hat{\lambda}$ calibrated so added failures $\leq \alpha$ with probability $\geq 1 - \delta$.

FALL BACK

Never costs more than the released sampler; no certificate means the reference.

THANK YOU.

QUESTIONS?